



Cape Peninsula
University of Technology

FACULTY OF INFORMATICS & DESIGN

ADVANCED DIPLOMA in INFORMATION & COMMUNICATIONS TECHNOLOGY

Subject Guide

APPLICATIONS DEVELOPMENT THEORY 4

SUBJECT GUIDE FOR:	2026
COURSE CODE:	APT470S
NQF LEVEL:	7
PRE-REQUISITE SUBJECTS:	Applications Development Theory 3 or equivalent

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1. INTRODUCTION

Welcome to Applications Development Theory 4. This subject follows on from the diploma subject of the same name, where the foundation for software development was set. Applications Development Theory 4 focuses on the following core topics related to the design and development of high-quality software artifacts: Software Requirements Engineering, Software Design based on the ISO/IEC 12207-2017 and ISO/IEC 15288-2015 standards; Software Configuration Management; and Software Quality.

This Study Guide provides an overview of the course, its objectives, expected learning outcomes, and the mode of assessment used in this course. The relevant resources and important dates are also listed.

2. GENERAL INFORMATION

2.1 Contact Information

DESIGNATION	NAME	LOCATION	TELEPHONE NUMBER	E-MAIL ADDRESS
Subject Lecturer	Prof A Ajibesin	Engineering		ajibesina@cput.ac.za
Domain Coordinator	Dr. Richard Maliwatu	Engineering 2.5	(021) 460 3010	maliwatur@cput.ac.za
Head of Department	Dr. T Ncubukezi	Engineering 2.30	(021) 460-3713	NcubukeziT@cput.ac.za
Secretary	N. Allie	Engineering 2.1	(021) 460-3010	allien@cput.ac.za

2.2 Class Times and Venues

As published on <http://myclassroom.cput.ac.za> and notice boards. This subject consists of a period per week.

2.3 Learner Management System (LMS)

Blackboard is used for general communications and announcements, posting of course material, and assignments. It is the learner's responsibility to check Blackboard regularly at <https://myclassroom.cput.ac.za> to remain up to date with all subject matter. Notifications will also be sent from Blackboard to the university's official email accounts for students at their **mycput email addresses**.

2.4 Queries (CRIMS System)

The CRIMS system is an online platform that helps students open tickets and raise subject- and other academic-related queries. Students must not open a CRIMS ticket if the matter can be addressed directly with the Lecturer/Subject Coordinator. You can access this system at <https://enquiries.cput.ac.za>

2.5 Recognition of Prior Learning (RPL)

In a situation where there is a claim of previous learning that equates to a qualification in Information Technology, the recognition of such a qualification will be at the discretion of the higher education institution. A structured process for the assessment of individual learners will be followed, on a case-by-case basis, vis-à-vis the exit-level outcomes of the qualification. Independent assessors will moderate this process and the assessment of individual cases.

In this course, we recognize RPL by granting credit to learners who demonstrate both theoretical and practical knowledge of software engineering and the development of large, software-intensive systems. This is not to be confused with an exemption granted for a similar subject studied at another recognized tertiary institution.

3 SUBJECT INFORMATION

3.1 Subject Credits

This subject has 20 credits at level 7 on the NQF. This means that the average student will spend 200 hours per year mastering the subject's outcomes. Of this, 30% will be contact hours, and the remainder will be self-study, assignments, tests, and other tasks. For further information on this, please refer to the Faculty Prospectus at

<http://www.cput.ac.za/academic/faculties/informaticsdesign/prospectus>

3.2 Length of the course

METHOD	NOTIONAL HOURS (1 Notional hour = 10 hours)
Formal Lectures	6
Practical / Assignments	4
Self-Study	8
Other	2
TOTAL	20

3.3 Learning Outcomes

The learning outcomes associated with this subject include the ability to –

- Recognise, articulate, and compare various models in advanced software development
- Understand and make comparisons among advanced techniques in the design of software systems
- Identify and compare tools required in effective software development and maintenance
- Effectively communicate and evaluate their work and that of their peers

3.4 Lecture Delivery Methods

The mode of lecture delivery will be based on the following: online lectures, mandatory and other readings, and assignments (Group and individual).

3.5 Prescribed Textbooks

Software Engineering, by Ian Sommerville 10th edition, Pearson Education Limited, 2016.

and

Software Engineering – A Practitioner's Approach 8th edition Roger Pressman McGraw Hill Education (2015).

3.6 Library and Readings

Additional reading is an essential part of the course, particularly for topics not covered in the prescribed textbooks. Students are expected to make use of the library and the Internet as additional sources of information.

4. ASSESSMENT

4.1 Assessment policy and regulations

This subject has four assessments. Assessments are a combination of Blackboard online tests, Blackboard quizzes, and assignments. Assignments are administered via Blackboard.

Test queries: The last day for any test queries is one week after the test was discussed in class. Thereafter, the test mark cannot be changed.

Absence from a test: If you are absent from a test, a valid medical certificate must be submitted to the faculty Customer Relationship Information Management system (CRIMs) at <https://enquiries.cput.ac.za/>. This should be done as soon as possible, preferably before the test and in line with the University regulations. You must also inform your lecturer on or before the day of the test that you are ill.

Due dates must be strictly adhered to. No late submissions are allowed for any subject in this Advanced Diploma.

4.2 Assessment Opportunities:

There will be one test per term. An Exception/Sick test will be carried out at the end of the semester. Students may qualify for one Exception/Sick test per subject per year.

4.3 Assessment Breakdown and Weight

ASSESSMENT BREAKDOWN AND WEIGHTS					
ASSESSMENT CRITERIA	1st TERM	2nd TERM	3rd TERM	4th TERM	TOTALS
Nature of assessment	Class Tests / Assignments	Written Exam	Class Tests / Assignments	Written Exam	100%
Will assessment be included in this term?	Yes	Yes	Yes	Yes	
Total assessment weight for the term	10%	30%	30%	30%	
Will the assessment be moderated?	No	Yes	No	Yes	

4.4 Tests and

Assignment Guidelines

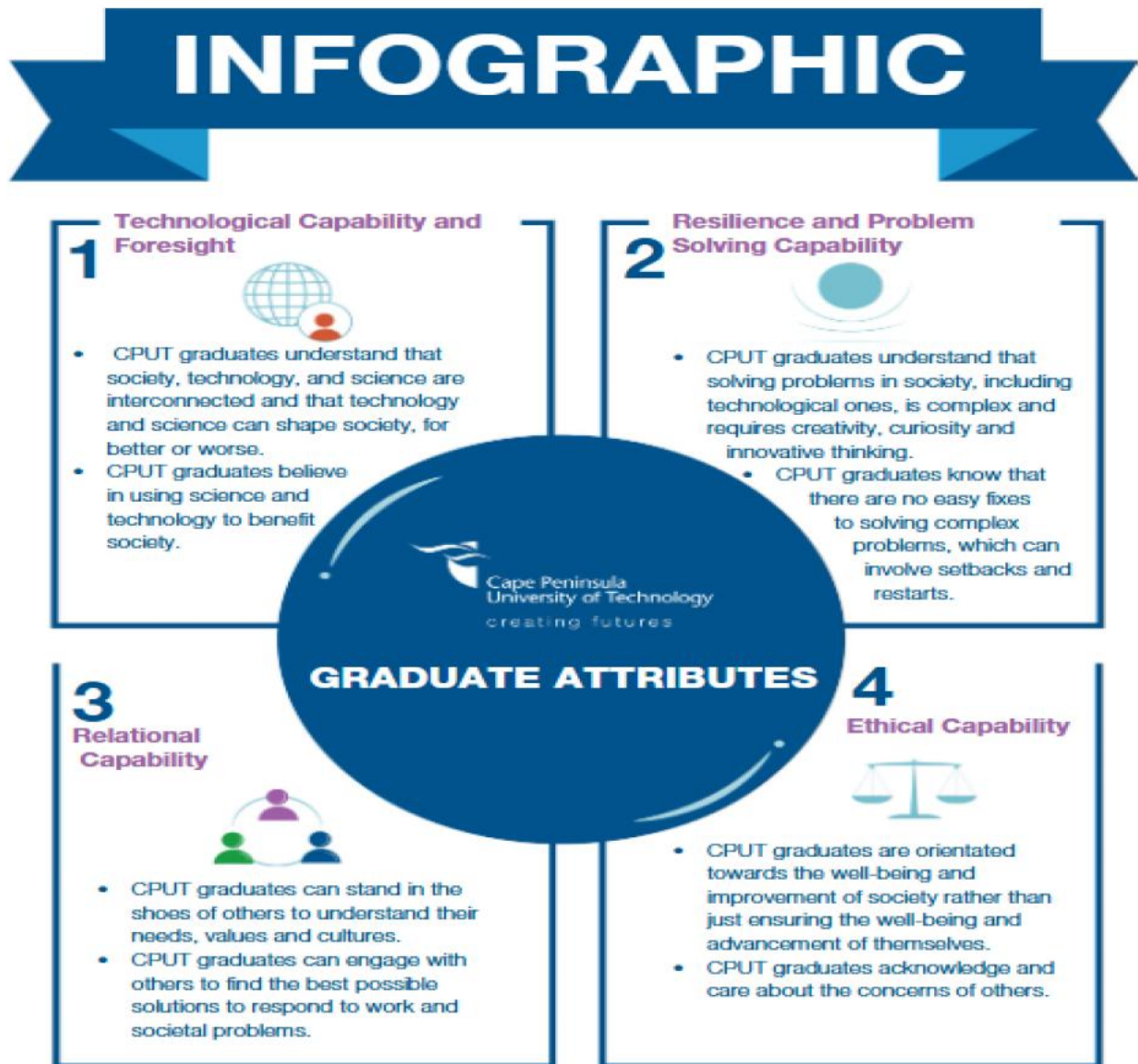
All tests are individual assessments. Assignments may either be done individually or as part of a group. A maximum of 4 students is allowed per group. All work must be carried out in a way that ensures each member contributes. The assignments are detailed in the Assignment Guide on Blackboard. Submission deadlines must be adhered to. No late submissions will be accepted.

5 TOPIC ORGANIZATION and SCHEDULE

No	Topic	Week of
Term 1	THEME 1: THE SOFTWARE PROCESS and REQUIREMENTS ENGINEERING	
1	The nature of software and Introduction to Software Engineering	02/02/2026
2	Software Processes and Models	09/02/2026
3	Agile Development	16/02/2026
4	Human Aspects of Software Engineering	3/02/2026
5	Requirements Engineering	02/03/2026
	Class Tests and Assignments for Term 1	
Term 2	THEME 2: MODELLING - SYSTEMS DESIGN and DEVELOPMENT	
6	Requirements Modelling	23/03/2026
7	Design Concepts	06/04/2026
8	Architectural Design	20/04/2026
9	Systems Design – component, UI, webApp design, mobileApp design,	04/05/2026
10	Reuse	11/05/2026
	Assignments, Class Tests and Exam on Semester 1 work	
Term 3	THEME 3: QUALITY MANAGEMENT	
11	Quality Concepts,	13/07/2026
12	Review Techniques	10/08/2026
13	Software Quality Assurance, Software Testing Strategies	17/08/2026
14	Software Testing – traditional, OO, webApp, mobileApp testing	24/08/2026
	Class Tests on Term 3 work	
Term 4	THEME 4: SECURITY ENGINEERING, SOFTWARE DEPENDABILITY, PROJECT MANAGEMENT and TRENDS	
15	Security Engineering	14/09/2026

16	Software Configuration Management	05/10/2026
17	Project – management, planning, configuration, risk management and process improvement	12/10/2026
18	Emerging Trends in Software Engineering	19/10/2026
Class Tests on Term 4 work and Exam on Semester 2 work		

6. GRADUATE ATTRIBUTES



6. STUDENT ACKNOWLEDGEMENT

CAPE PENINSULA UNIVERSITY OF TECHNOLOGY STUDENT ACKNOWLEDGEMENT

2026

ADVANCED DIPLOMA: INFORMATION and COMMUNICATIONS TECHNOLOGY APPLICATIONS DEVELOPMENT THEORY 4

ACKNOWLEDGE THAT YOU HAVE READ THE SUBJECT GUIDE AND UNDERSTAND ITS CONTENT

Please read the following statement and complete the spaces that follow. Make a copy of this page for your own reference. Hand-in this original page during the next class session.

I have read the subject guide of **APPLICATIONS DEVELOPMENT THEORY 4** and understand the information and my responsibilities as a learner for this subject. I am also aware that the Subject Guide may be revised due to unforeseen circumstances and it remains my responsibility to adhere to the most recent guide.

Name & Surname: (Print)

Student Number:

Contact number: (home/res) Contact number: (cell)

Signature:

Date: